

Online algorithm

From Wikipedia, the free encyclopedia

In computer science, an **online algorithm** is one that can process its input piece-by-piece in a serial fashion, i.e., in the order that the input is fed to the algorithm, without having the entire input available from the start.

In contrast, an **offline algorithm** is given the whole problem data from the beginning and is required to output an answer which solves the problem at hand.

Some "offline vs online" algorithms:

Category	offline	online	Description
Sorting:	Selection sort	Insertion sort	<i>Selection sort</i> requires that the entire list be given before it can sort it, while <i>Insertion sort</i> doesn't.
XML parsers:	DOM	SAX	<i>SAX parsers</i> operate on each piece of the XML document sequentially, <i>DOM parsers</i> operates on the document as a whole.

Not every *online algorithm* has an *offline* counterpart.

Contents

- 1 Explaining
 - 1.1 Other interpretations
 - 1.2 Examples
- 2 Online problems redirects here
- 3 See also
- 4 References
- 5 External links

Explaining

Because it does not know the whole input, an online algorithm is forced to make decisions that may later turn out not to be optimal, and the study of online algorithms has focused on the quality of decision-making that is possible in this setting. Competitive analysis formalizes this idea by comparing the relative performance of an online and offline algorithm for the same problem instance. Specifically, the competitive ratio of an algorithm, is defined as the worst-case ratio of its cost divided by the optimal cost, over all possible inputs. The competitive ratio of an online problem is the best competitive ratio achieved by an online algorithm. Intuitively, the competitive ratio of an algorithm gives a measure on the quality of solutions produced by this algorithm, while the competitive ratio of a problem shows the importance of knowing the future for this problem.

Other interpretations

For other points of view on *online inputs to algorithms*, see

- streaming algorithm: focusing on the amount of memory needed to accurately represent past inputs;
- dynamic algorithm: focusing on the time complexity of maintaining solutions to problems with online inputs.

Examples

Some *online algorithms*:

- Insertion sort
- Perceptron
- Reservoir sampling
- Greedy algorithm
- Adversary model
- Metrical task systems
- Odds algorithm
- Page replacement algorithm
- Algorithms for calculating variance
- Ukkonen's algorithm

Online problems redirects here

A problem exemplifying the concepts of online algorithms is the Canadian Traveller Problem. The goal of this problem is to minimize the cost of reaching a target in a weighted graph where some of the edges are unreliable and may have been removed from the graph. However, that an edge has been removed (*failed*) is only revealed to *the traveller* when she/he reaches one of the edge's endpoints. The worst case for this problem is simply that all of the unreliable edges fail and the problem reduces to the usual Shortest Path Problem. An alternative analysis of the problem can be made with the help of competitive analysis. For this method of analysis, the offline algorithm knows in advance which edges will fail and the goal is to minimize the ratio between the online and offline algorithms' performance. This problem is PSPACE-complete.

There are many formal problems that offers more than one *online algorithm* as solution:

- K-server problem
- Job shop scheduling problem
- List update problem
- Bandit problem
- Secretary problem
- Search games
- Ski rental problem
- Linear search problem

See also

- dynamic algorithm
- Streaming algorithm
- Real-time computing
- Sequential algorithm
- Online machine learning/Offline learning

References

- Borodin, A.; El-Yaniv, R. (1998). *Online Computation and Competitive Analysis* (<http://www.cs.technion.ac.il/~rani/book.html>). Cambridge University Press. ISBN 0-521-56392-5.

External links

- Bibliography of papers on online algorithms (<http://www.cs.ucr.edu/~marek/pubs/online.bib>)

Retrieved from "https://en.wikipedia.org/w/index.php?title=Online_algorithm&oldid=671141916"

Categories: Online algorithms

-
- This page was last modified on 12 July 2015, at 18:48.
 - Text is available under the Creative Commons Attribution-ShareAlike License; additional terms may apply. By using this site, you agree to the Terms of Use and Privacy Policy. Wikipedia® is a registered trademark of the Wikimedia Foundation, Inc., a non-profit organization.